

COMPLEX CONJUGATES SERIES

Exercise 2.4 Solved Masterclass Solutions

Rigorous step-by-step intermediate solutions for Question 1 to Question 7, with parallel evaluations and continuous overbar conjugate notations.

Q1(i): Conjugate of a Complex Sum

Simplify the linear sum and apply the continuous conjugate overbar transformation:

Step 1: Parallel Sum

Express the combined expression before applying conjugation properties:

$$z = (5 + 9i) + (2 - 4i)$$

Add real components and imaginary coefficients parallelly:

$$z = (5 + 2) + (9 - 4)i \quad z = 7 + 5i$$

Step 2: Conjugation

Evaluate the overbar spanning over the entire sum term:

$$\overline{(5 + 9i) + (2 - 4i)} = \overline{7 + 5i}$$

Swap the sign of the imaginary coefficient:

$$= 7 - 5i$$

Matches standard rectangular form $x + iy$ where $x=7$ and $y=-5$.

Q1(ii): Division Rationalization

Convert the complex division quotient fraction into standard rectangular form:

÷ The Problem

$$z = \frac{10 - 5i}{6 + 2i}$$

We rationalize by multiplying both top and bottom with the denominator's conjugate.

🔄 Step 1: Conjugate Product

The conjugate of the denominator is:

$$\overline{6 + 2i} = 6 - 2i$$

Multiply terms parallelly:

$$z = \frac{(10 - 5i)(6 - 2i)}{(6 + 2i)(6 - 2i)}$$

Q1(ii): Evaluating Products

Evaluating numerator and denominator expansions simultaneously:

↑ Numerator & Denominator

Expand top binomial elements:

$$(10 - 5i)(6 - 2i) = 60 - 20i - 30i + 10i^2$$

Since $i^2 = -1$: $= 50 - 50i$

Apply sum of squares for denominator:

$$(6)^2 + (2)^2 = 36 + 4 = 40$$

✓ Final Simplification

Divide the parallel resulting components:

$$z = \frac{50 - 50i}{40} = \frac{50}{40} - \frac{50}{40}i$$

Reduce standard coefficients:

$$z = \frac{5}{4} - \frac{5}{4}i$$

Q1(iii): Linear Conjugate Fraction

Find the rectangular form of the combined conjugate and reciprocal terms:

Problem Statement

$$z = \overline{3i} + \frac{1}{2 - i}$$

Observe the individual component conjugate evaluation first:

$$\overline{3i} = \overline{0 + 3i} = -3i$$

Step 1: Second Term

Rationalize the rational part by $(2 + i)$:

$$\frac{1}{2 - i} = \frac{1(2 + i)}{(2 - i)(2 + i)}$$

$$= \frac{2 + i}{4 + 1} = \frac{2 + i}{5} = \frac{2}{5} + \frac{1}{5}i$$

Q1(iii): Combining the Terms

Sum the evaluated real and imaginary parts parallelly:

Step 2: Linear Sum

Substitute evaluated components back into the formula:

$$z = -3i + \left(\frac{2}{5} + \frac{1}{5}i\right)$$

Isolate real and group imaginary parts:

$$z = \frac{2}{5} + \left(\frac{1}{5} - 3\right)i$$

Final Solution

Take common denominators for the imaginary part:

$$\frac{1}{5} - \frac{15}{5} = -\frac{14}{5}$$

Standard rectangular coordinate expression is:

$$z = \frac{2}{5} - \frac{14}{5}i$$

Q2(i): Real Part of Reciprocal

Let $z = x + iy$. Determine the real part of reciprocal $Re(\frac{1}{z})$:

Step 1: Inverse Setup

Express the reciprocal of the complex variable:

$$\frac{1}{z} = \frac{1}{x + iy}$$

Multiply top and bottom by conjugate $(x - iy)$:

$$\frac{1}{z} = \frac{1(x - iy)}{(x + iy)(x - iy)}$$

Step 2: Evaluation

Apply the sum of squares identity to the denominator:

$$(x + iy)(x - iy) = x^2 + y^2$$

Assembling the parallel fractions:

$$\frac{1}{z} = \frac{x - iy}{x^2 + y^2}$$

Q2(i): Extracting Real Part

Split the ratio and isolate the scalar real coefficient:

Step 3: Splitting Fractions

Rewrite the fraction coordinate components separately:

$$\frac{1}{z} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$$

This matches standard rectangular coordinate format:

$$A + iB$$

Final Components

Real Part $\text{Re}(1/z)$:

$$\text{Re}\left(\frac{1}{z}\right) = \frac{x}{x^2 + y^2}$$

Imaginary Part $\text{Im}(1/z)$:

$$\text{Im}\left(\frac{1}{z}\right) = -\frac{y}{x^2 + y^2}$$

Q2(ii): Real Part of Product

Let $z = x + iy$. Determine the real part of imaginary conjugate product $Re(i\bar{z})$:

✂ Step 1: Conjugate

Substitute the overbar conjugate representation:

$$\bar{z} = x - iy$$

Multiply this term parallelly by i :

$$i\bar{z} = i(x - iy)$$

🧮 Step 2: Multiplication

Distribute imaginary coefficient:

$$i\bar{z} = ix - i^2y$$

Substitute $i^2 = -1$:

$$i\bar{z} = ix - (-1)y = y + ix$$

Isolate standard real component:

$$Re(i\bar{z}) = y$$

Q2(iii): Imaginary Part of Function

Let $z = x + iy$. Determine the imaginary coefficient $Im(3z + 4\bar{z} - 4i)$:

Step 1: Expression Expansion

Substitute variables into the polynomial sum:

$$W = 3(x + iy) + 4(x - iy) - 4i$$

Expand terms parallelly:

$$W = 3x + i3y + 4x - i4y - 4i$$

Step 2: Like-Terms

Group real and imaginary coefficients:

$$W = (3x + 4x) + i(3y - 4y - 4) \quad W = 7x + i(-y - 4)$$

Extract imaginary coefficient:

$$Im(W) = -y - 4$$

Q3: Conjugate Product Base

Given $z1 = 2 - i$ and $z2 = -4 + 3i$. First, compute their product $z1z2$:

Step 1: Expansion

Multiply binomial elements:

$$z1z2 = (2 - i)(-4 + 3i)$$

Apply distributive rule parallelly:

$$= 2(-4 + 3i) - i(-4 + 3i) = -8 + 6i + 4i - 3i^2$$

Step 2: Collection

Substitute imaginary constant value:

$$= -8 + 10i - 3(-1) = -8 + 10i + 3$$

Resulting complex product base:

$$z1z2 = -5 + 10i$$

Q3: Inverse of the Product

Find the multiplicative inverse of the evaluated complex product:

Step 3: Inverse Setup

Express the reciprocal of the product:

$$(z_1 z_2)^{-1} = \frac{1}{-5 + 10i}$$

First factor out scalar 5 from the denominator:

$$= \frac{1}{5(-1 + 2i)}$$

Final Rationalization

Multiply top and bottom elements by conjugate $(-1 - 2i)$:

$$= \frac{-1 - 2i}{5((-1)^2 + (2)^2)} = \frac{-1 - 2i}{5(1 + 4)}$$

$$(z_1 z_2)^{-1} = -\frac{1}{25} - \frac{2}{25}i$$

Q3: Complex Ratio Division

Find the rectangular form coordinate of quotient z_1/z_2 :

÷ Quotient Setup

$$\frac{z_1}{z_2} = \frac{2 - i}{-4 + 3i}$$

Multiply both components by conjugate $(-4 - 3i)$:

$$\frac{z_1}{z_2} = \frac{(2 - i)(-4 - 3i)}{(-4 + 3i)(-4 - 3i)}$$

🧮 Step-by-Step Multiplication

Evaluate expansions parallelly:

$$\text{Numerator} = -8 - 6i + 4i + 3i^2 = -11 - 2i$$

$$\text{Denominator} = (-4)^2 + (3)^2 = 16 + 9 = 25$$

Resulting rectangular fraction:

$$\frac{z_1}{z_2} = -\frac{11}{25} - \frac{2}{25}i$$

Q4: Inverse Parallel Formula

The complex variables are related by $\frac{1}{u} = \frac{1}{v} + \frac{1}{w}$. If $v = 3 - 4i$ and $w = 4 + 3i$, find u :

Step 1: Parallel Reciprocals

Solve algebraically for the common fraction denominator:

$$\frac{1}{u} = \frac{v + w}{vw} \Rightarrow u = \frac{vw}{v + w}$$

Evaluate numerator product and denominator sum parallelly.

Step 2: Components

Denominator Sum:

$$v + w = (3 - 4i) + (4 + 3i) = 7 - i$$

Numerator Product:

$$vw = (3 - 4i)(4 + 3i) = 12 + 9i - 16i - 12i^2$$

$$vw = 12 - 7i + 12 = 24 - 7i$$

Q4: Solving for u

Assemble coordinates and execute division rationalization:

÷ Quotient Setup

$$u = \frac{24 - 7i}{7 - i}$$

Multiply terms by conjugate factor $(7 + i)$:

$$u = \frac{(24 - 7i)(7 + i)}{(7 - i)(7 + i)}$$

✓ Final Calculation

Expand products parallelly:

$$\text{Numerator} = 168 + 24i - 49i - 7i^2 = 175 - 25i$$

$$\text{Denominator} = (7)^2 + (1)^2 = 49 + 1 = 50$$

Simplify quotients:

$$u = \frac{175 - 25i}{50} = \frac{175}{50} - \frac{25}{50}i$$

$$u = \frac{7}{2} - \frac{1}{2}i$$

Q5(i): Conjugate Real Identity Proof

Prove that complex number z is purely real if and only if $z = \bar{z}$:

↻ 1. Direct Proof

Assume $z = x + iy$ is purely real, meaning $y = 0$:

$$z = x + i(0) = x$$

Take conjugate overbar over both parts:

$$\bar{z} = \bar{x} = x$$

Therefore, $z = \bar{z}$ holds true.

↔ 2. Converse Proof

Assume $z = \bar{z}$ where $z = x + iy$:

$$x + iy = x - iy$$

Subtract x and group terms parallelly:

$$iy = -iy \Rightarrow 2iy = 0 \Rightarrow y = 0$$

Since the imaginary coefficient is zero, z is purely real. Q.E.D.

Q5(ii): Real & Imaginary Parts Proof

Prove that $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$ and $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$:

1. Proof for $\operatorname{Re}(z)$

Let $z = x + iy$ and $\bar{z} = x - iy$:

$$z + \bar{z} = (x + iy) + (x - iy)$$

$$z + \bar{z} = 2x \quad \Rightarrow \quad \frac{z + \bar{z}}{2} = x$$

Since $\operatorname{Re}(z) = x$, this verifies the first property.

2. Proof for $\operatorname{Im}(z)$

Evaluate the coordinate difference parallelly:

$$z - \bar{z} = (x + iy) - (x - iy)$$

$$z - \bar{z} = 2iy \quad \Rightarrow \quad \frac{z - \bar{z}}{2i} = y$$

Since $\operatorname{Im}(z) = y$, this verifies the second property. Q.E.D.

Q6: Polar Exponential Setup

Find the least positive integer n for which the expression $(\sqrt{3} + i)^n$ is: (i) real, and (ii) purely imaginary.

Step 1: Converting to Polar Form

Let the complex variable be $z = \sqrt{3} + i$. Identify components:

Real coordinate $x = \sqrt{3}$, Imaginary coordinate $y = 1$.

Compute Modulus (r):

$$r = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{3 + 1} = \sqrt{4} = 2$$

Compute Phase Angle (θ):

$$\theta = \tan^{-1}(1 / \sqrt{3}) = \pi / 6$$

Step 2: Applying De Moivre's Theorem

Express the complex variable z in Polar form:

$$z = 2 * (\cos(\pi / 6) + i * \sin(\pi / 6))$$

Raise to the power n using De Moivre's rule:

$$z^n = 2^n * (\cos(n\pi / 6) + i * \sin(n\pi / 6))$$

-  For **purely real** values: imaginary coefficient $\sin(n\pi / 6) = 0$.
-  For **purely imaginary** values: real coefficient $\cos(n\pi / 6) = 0$.

Q6: Evaluating Least Integer n

Solve the trigonometric equations to find the minimum positive integer n:

Case (i): Purely Real Value

The imaginary coefficient must resolve to zero:

$$\sin(n\pi / 6) = 0$$

General solutions occur at integer multiples of Pi:

$$n\pi / 6 = k * \pi \text{ (where } k \text{ is an integer)}$$

Simplify the relationship to solve for n:

$$n = 6 * k$$

To find the smallest positive integer n, choose k = 1:

$$n = 6$$

Case (ii): Purely Imaginary

The real coefficient must resolve to zero:

$$\cos(n\pi / 6) = 0$$

General solutions occur at odd half-multiples of Pi:

$$n\pi / 6 = (2k + 1) * (\pi / 2) \text{ (where } k \text{ is an integer)}$$

Simplify the relationship to solve for n:

$$n = 3 * (2k + 1)$$

To find the smallest positive integer n, choose k = 0:

$$n = 3$$

Q7(i): Conjugate Symmetry Proof

Show that $z = (2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}$ is purely imaginary using overbars:

Proof Strategy

Recall the purely imaginary condition property from Slide 19:

$$\overline{\overline{z}} = -z$$

Apply the continuous conjugate overbar over the subtraction difference:

$$\overline{z} = \overline{(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}}$$

Proof Completion

Distribute over addition and swap the power exponent:

$$\begin{aligned}\overline{z} &= \overline{(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}} = \overline{(2 + i\sqrt{3})^{10}} - \overline{(2 - i\sqrt{3})^{10}} \\ &= (2 - i\sqrt{3})^{10} - (2 + i\sqrt{3})^{10} \\ &= -[(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}] = -z\end{aligned}$$

Since conjugate of z equals negative z , the term is purely Imaginary. Proved.

Q7(ii): Double Rationalization Base

Show that the exponent sum expression is purely real. First, simplify internal bases parallelly:

Step 1: Base 1

Rationalize the first term fraction by conjugate $(9 - i)$:

$$\frac{19 - 7i}{9 + i} = \frac{(19 - 7i)(9 - i)}{9^2 + 1^2}$$

$$\text{Numerator} = 171 - 19i - 63i + 7i^2 = 164 - 82i$$

$$\text{Value} = \frac{164 - 82i}{82} = 2 - i$$

Step 2: Base 2

Rationalize the second term fraction by conjugate $(7 + 6i)$:

$$\frac{20 - 5i}{7 - 6i} = \frac{(20 - 5i)(7 + 6i)}{7^2 + 6^2}$$

$$\text{Numerator} = 140 + 120i - 35i - 30i^2 = 170 + 85i$$

$$\text{Value} = \frac{170 + 85i}{85} = 2 + i$$

Q7(ii): Exponent Real Proof

Substituting evaluated bases back to finalize the overbar conjugate proof:

✓ Step 3: Expression

Express the simplified term exponent sum:

$$z = (2 - i)^{12} + (2 + i)^{12}$$

Apply the continuous overbar conjugate over the whole sum:

$$\overline{z} = \overline{(2 - i)^{12} + (2 + i)^{12}}$$

✓ Proof Completion

Distribute conjugation and swap exponent powers:

$$\begin{aligned}\overline{z} &= \overline{(2 - i)^{12}} + \overline{(2 + i)^{12}} = \overline{(2 - i)^{12}} + \overline{(2 + i)^{12}} \\ &= (2 + i)^{12} + (2 - i)^{12} = z\end{aligned}$$

Since conjugate of z equals z , the term evaluates to a purely Real value. Proved.